

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (AI)

DEEP LEARNING LAB RECORD

LAB - 01

Student Details

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Course Name: Deep Learning

Course Code: 22AIE304

Experiment Title

Implementation of Perceptron for Logic Gates (AND, OR, XOR) and Deep Neural Network (DNN) using TensorFlow on the MNIST Dataset

Aim

To implement a single-layer perceptron for AND and OR logic gates, implement a multilayer perceptron for the XOR logic gate, and build a Deep Neural Network (DNN) using TensorFlow to classify handwritten digits from the MNIST dataset.

Objectives

- Understand the working of a Perceptron.
 - Implement AND and OR gates using a single-layer perceptron.
 - Implement XOR gate using a Multi-Layer Perceptron (MLP).
 - Build a Deep Neural Network using TensorFlow.
 - Train and evaluate the model using the MNIST handwritten digit dataset.
-

Software Requirements

- Python 3.x

- NumPy
 - Scikit-Learn
 - TensorFlow
 - Torch (Optional)
 - TorchVision (Optional)
 - Jupyter Notebook / Anaconda
-

Outcome

- Successfully implemented AND and OR gates using a Perceptron.
 - Successfully implemented XOR gate using a Multi-Layer Perceptron.
 - Successfully trained a Deep Neural Network on the MNIST dataset.
 - Achieved approximately **97% Test Accuracy** on the MNIST dataset.
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Result

The objectives of the experiment were successfully achieved. The Perceptron correctly predicted the outputs for AND and OR logic gates, the Multi-Layer Perceptron successfully learned the XOR function, and the Deep Neural Network achieved high classification accuracy on the MNIST handwritten digit dataset.

In [1]:

```
# Install all required libraries  
!pip install numpy scikit-learn tensorflow torch torchvision
```

Requirement already satisfied: numpy in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (1.26.4)
Requirement already satisfied: scikit-learn in c:\programdata\anaconda3\lib\site-packages (0.24.2)
Requirement already satisfied: tensorflow in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (2.20.0)
Collecting torch
Using cached torch-2.8.0-cp39-cp39-win_amd64.whl (241.2 MB)
Collecting torchvision
Using cached torchvision-0.23.0-cp39-cp39-win_amd64.whl (1.6 MB)
Requirement already satisfied: threadpoolctl>=2.0.0 in c:\programdata\anaconda3\lib\site-packages (from scikit-learn) (2.2.0)
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Requirement already satisfied: joblib>=0.11 in c:\programdata\anaconda3\lib\site-packages (from scikit-learn) (1.1.0)
Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (0.5.4)
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Requirement already satisfied: astunparse>=1.6.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (1.6.3)
Requirement already satisfied: typing_extensions>=3.6.6 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (4.15.0)
Requirement already satisfied: h5py>=3.11.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (3.14.0)
Requirement already satisfied: keras>=3.10.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (3.10.0)
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Requirement already satisfied: six>=1.12.0 in c:\programdata\anaconda3\lib\site-packages (from tensorflow) (1.16.0)
Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (0.7.0)
Requirement already satisfied: tensorboard~2.20.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (2.20.0)
Requirement already satisfied: protobuf>=5.28.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (6.33.6)
Requirement already satisfied: requests<3,>=2.21.0 in c:\programdata\anaconda3\lib\site-packages (from tensorflow) (2.26.0)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (1.80.0)
Requirement already satisfied: flatbuffers>=24.3.25 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (25.12.19)
Requirement already satisfied: termcolor>=1.1.0 in c:\users\ch.sc.u4aie24057

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\appdata\roaming\python\python39\site-packages (from tensorflow) (3.1.0)
Collecting sympy>=1.13.3
  Using cached sympy-1.14.0-py3-none-any.whl (6.3 MB)
Requirement already satisfied: fsspec in c:\programdata\anaconda3\lib\site-packages (from torch) (2021.10.1)
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Requirement already satisfied: jinja2 in c:\programdata\anaconda3\lib\site-packages (from torch) (2.11.3)
Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in c:\programdata\anaconda3\lib\site-packages (from torchvision) (8.4.0)
Requirement already satisfied: wheel<1.0,>=0.23.0 in c:\programdata\anaconda3\lib\site-packages (from astunparse>=1.6.0->tensorflow) (0.37.0)
Requirement already satisfied: namex in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from keras>=3.10.0->tensorflow) (0.1.0)
Requirement already satisfied: optree in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from keras>=3.10.0->tensorflow) (0.19.1)
Requirement already satisfied: rich in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from keras>=3.10.0->tensorflow) (15.0.0)
Requirement already satisfied: urllib3<1.27,>=1.21.1 in c:\programdata\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (1.26.7)
Requirement already satisfied: charset-normalizer~=2.0.0 in c:\programdata\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (2.0.4)
Requirement already satisfied: certifi>=2017.4.17 in c:\programdata\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (2021.10.8)
Requirement already satisfied: idna<4,>=2.5 in c:\programdata\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (3.2)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in c:\programdata\anaconda3\lib\site-packages (from sympy>=1.13.3->torch) (1.2.1)
Requirement already satisfied: markdown>=2.6.8 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorboard~=2.20.0->tensorflow) (3.9)
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Requirement already satisfied: zipp>=0.5 in c:\programdata\anaconda3\lib\site-packages (from importlib-metadata>=4.4->markdown>=2.6.8->tensorboard~=2.20.0->tensorflow) (3.6.0)
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Requirement already satisfied: mdurl~=0.1 in c:\programdata\anaconda3\lib\site-packages (from markdown-it-py>=2.2.0->rich->keras>=3.10.0->tensorflow) (0.1.2)
Installing collected packages: sympy, torch, torchvision
  Attempting uninstall: sympy
    Found existing installation: sympy 1.9
    Uninstalling sympy-1.9:
ERROR: Could not install packages due to an OSError: [WinError 5] Access is denied: 'c:\\programdata\\anaconda3\\lib\\site-packages\\__pycache__\\isympy.c

```

python-39.pyc'

Consider using the `--user` option or check the permissions.

In [2]:

```
import numpy as np

# Define Step Activation Function
def step_function(x):
    return 1 if x >= 0 else 0

# Perceptron Predict Function
def perceptron(x, weights, bias):
    linear_output = np.dot(x, weights) + bias
    return step_function(linear_output)

# Inputs for 2-input logic gates
inputs = np.array([[0, 0], [0, 1], [1, 0], [1, 1]])

# --- AND GATE CONFIGURATION ---
# Weights and bias that satisfy the AND logic
weights_and = np.array([1, 1])
bias_and = -1.5

print("--- AND Gate Predictions ---")
for x in inputs:
    print(f"Input: {x} -> Predicted Output: {perceptron(x, weights_and, bias_and)}")

# --- OR GATE CONFIGURATION ---
# Weights and bias that satisfy the OR logic
weights_or = np.array([1, 1])
bias_or = -0.5

print("\n--- OR Gate Predictions ---")
for x in inputs:
    print(f"Input: {x} -> Predicted Output: {perceptron(x, weights_or, bias_or)}")

--- AND Gate Predictions ---
Input: [0 0] -> Predicted Output: 0
Input: [0 1] -> Predicted Output: 0
Input: [1 0] -> Predicted Output: 0
Input: [1 1] -> Predicted Output: 1

--- OR Gate Predictions ---
Input: [0 0] -> Predicted Output: 0
Input: [0 1] -> Predicted Output: 1
Input: [1 0] -> Predicted Output: 1
Input: [1 1] -> Predicted Output: 1
```

In [3]:

```
import numpy as np
from sklearn.neural_network import MLPClassifier

# XOR inputs and true outputs
X = np.array([[0, 0], [0, 1], [1, 0], [1, 1]])
y = np.array([0, 1, 1, 0])

# Switched solver to 'adam' and adjusted learning rate for stable convergence
mlp = MLPClassifier(
    hidden_layer_sizes=(4, 4), # Using two small hidden layers makes it high
    activation='relu',
    solver='adam',
    learning_rate_init=0.01,
    max_iter=2000,
    random_state=42
```

```

)

# Train the model
mlp.fit(X, y)

# Test the model
print("--- XOR Gate Predictions ---")
for x in X:
    pred = mlp.predict([x])
    print(f"Input: {x} -> Predicted Output: {pred[0]}")

--- XOR Gate Predictions ---
Input: [0 0] -> Predicted Output: 0
Input: [0 1] -> Predicted Output: 1
Input: [1 0] -> Predicted Output: 1
Input: [1 1] -> Predicted Output: 0

```

In [4]:

```

import sys
!{sys.executable} -m pip install tensorflow

```

```

Requirement already satisfied: tensorflow in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (2.20.0)
Requirement already satisfied: typing_extensions>=3.6.6 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (4.15.0)
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Requirement already satisfied: setuptools in c:\programdata\anaconda3\lib\site-packages (from tensorflow) (58.0.4)
Requirement already satisfied: termcolor>=1.1.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (3.1.0)
Requirement already satisfied: six>=1.12.0 in c:\programdata\anaconda3\lib\site-packages (from tensorflow) (1.16.0)
Requirement already satisfied: wrapt>=1.11.0 in c:\programdata\anaconda3\lib\site-packages (from tensorflow) (1.12.1)
Requirement already satisfied: keras>=3.10.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (3.10.0)
Requirement already satisfied: tensorboard~=2.20.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (2.20.0)
Requirement already satisfied: absl-py>=1.0.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (2.3.1)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (1.80.0)
Requirement already satisfied: google_pasta>=0.1.1 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (0.2.0)
Requirement already satisfied: packaging in c:\programdata\anaconda3\lib\site-packages (from tensorflow) (21.0)
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Requirement already satisfied: libclang>=13.0.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (18.1.1)
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Requirement already satisfied: flatbuffers>=24.3.25 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (25.12.19)

Requirement already satisfied: protobuf>=5.28.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (6.33.6)

Requirement already satisfied: wheel<1.0,>=0.23.0 in c:\programdata\anaconda3\lib\site-packages (from astunparse>=1.6.0->tensorflow) (0.37.0)

Requirement already satisfied: namex in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from keras>=3.10.0->tensorflow) (0.1.0)

Requirement already satisfied: rich in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from keras>=3.10.0->tensorflow) (15.0.0)

Requirement already satisfied: optree in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from keras>=3.10.0->tensorflow) (0.19.1)

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Requirement already satisfied: urllib3<1.27,>=1.21.1 in c:\programdata\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (1.26.7)

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Requirement already satisfied: charset-normalizer~=2.0.0 in c:\programdata\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (2.0.4)

Requirement already satisfied: markdown>=2.6.8 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorboard~=2.20.0->tensorflow) (3.9)

Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorboard~=2.20.0->tensorflow) (0.7.2)

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Requirement already satisfied: werkzeug>=1.0.1 in c:\programdata\anaconda3\lib\site-packages (from tensorboard~=2.20.0->tensorflow) (2.0.2)

Requirement already satisfied: importlib-metadata>=4.4 in c:\programdata\anaconda3\lib\site-packages (from markdown>=2.6.8->tensorboard~=2.20.0->tensorflow) (4.8.1)

Requirement already satisfied: zipp>=0.5 in c:\programdata\anaconda3\lib\site-packages (from importlib-metadata>=4.4->markdown>=2.6.8->tensorboard~=2.20.0->tensorflow) (3.6.0)

Requirement already satisfied: pyparsing>=2.0.2 in c:\programdata\anaconda3\lib\site-packages (from packaging->tensorflow) (3.0.4)

Requirement already satisfied: pygments<3.0.0,>=2.13.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from rich->keras>=3.10.0->tensorflow) (2.20.0)

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In [5]:

```
pip install tensorflow
```

Requirement already satisfied: tensorflow in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (2.20.0)

Requirement already satisfied: astunparse>=1.6.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (1.6.3)

Requirement already satisfied: h5py>=3.11.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (3.14.0)

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Requirement already satisfied: grpcio<2.0,>=1.24.3 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (1.80.0)

Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (0.5.4)

Requirement already satisfied: opt_einsum>=2.3.2 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (3.4.0)

Requirement already satisfied: termcolor>=1.1.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (3.1.0)

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Requirement already satisfied: flatbuffers>=24.3.25 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (25.12.19)

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Requirement already satisfied: certifi>=2017.4.17 in c:\programdata\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (2021.10.8)

Requirement already satisfied: urllib3<1.27,>=1.21.1 in c:\programdata\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (1.26.7)

Requirement already satisfied: charset-normalizer~=2.0.0 in c:\programdata\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (2.0.4)

Requirement already satisfied: idna<4,>=2.5 in c:\programdata\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (3.2)

Requirement already satisfied: pillow in c:\programdata\anaconda3\lib\site-packages (from tensorboard~=2.20.0->tensorflow) (8.4.0)

Requirement already satisfied: werkzeug>=1.0.1 in c:\programdata\anaconda3\lib\site-packages (from tensorboard~=2.20.0->tensorflow) (2.0.2)

Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorboard~=2.20.0->tensorflow) (0.7.2)

Requirement already satisfied: markdown>=2.6.8 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorboard~=2.20.0->tensorflow) (3.9)

Requirement already satisfied: importlib-metadata>=4.4 in c:\programdata\anaconda3\lib\site-packages (from markdown>=2.6.8->tensorboard~=2.20.0->tensorflow)

w) (4.8.1)

Requirement already satisfied: zipp>=0.5 in c:\programdata\anaconda3\lib\site-packages (from importlib-metadata>=4.4->markdown>=2.6.8->tensorboard~>2.20.0->tensorflow) (3.6.0)

Requirement already satisfied: pyparsing>=2.0.2 in c:\programdata\anaconda3\lib\site-packages (from packaging->tensorflow) (3.0.4)

Requirement already satisfied: markdown-it-py>=2.2.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from rich->keras>=3.10.0->tensorflow) (3.0.0)

Requirement already satisfied: pygments<3.0.0,>=2.13.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from rich->keras>=3.10.0->tensorflow) (2.20.0)

Requirement already satisfied: mdurl~>0.1 in c:\programdata\anaconda3\lib\site-packages (from markdown-it-py>=2.2.0->rich->keras>=3.10.0->tensorflow) (0.1.2)

Note: you may need to restart the kernel to use updated packages.

In [6]:

```
import tensorflow as tf
from tensorflow.keras import layers, models

# 1. Load and preprocess the MNIST dataset
print("Loading MNIST dataset...")
mnist = tf.keras.datasets.mnist
(X_train, y_train), (X_test, y_test) = mnist.load_data()

# Normalize pixel values to be between 0 and 1
X_train, X_test = X_train / 255.0, X_test / 255.0

# 2. Build the Deep Neural Network (DNN) Architecture
print("Building the model architecture...")
model = models.Sequential([
    layers.Flatten(input_shape=(28, 28)),      # Flattens the 28x28 images into
    layers.Dense(128, activation='relu'),      # Hidden Layer 1 (128 neurons)
    layers.Dense(64, activation='relu'),       # Hidden Layer 2 (64 neurons)
    layers.Dense(10, activation='softmax')     # Output Layer (10 classes for
])

# 3. Compile the model
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

# 4. Train the model
print("Training the Deep Neural Network on MNIST...")
model.fit(X_train, y_train, epochs=5, batch_size=64)

# 5. Evaluate the model on test data
print("\nEvaluating model performance on test data...")
test_loss, test_acc = model.evaluate(X_test, y_test, verbose=2)
print(f"\nFinal Test Accuracy: {test_acc * 100:.2f}%")
```

Loading MNIST dataset...

Building the model architecture...

Training the Deep Neural Network on MNIST...

C:\Users\ch.sc.u4aie24057\AppData\Roaming\Python\Python39\site-packages\keras\src\layers\reshaping\flatten.py:37: UserWarning: Do not pass an `input\_shape` or `input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

super().__init__(**kwargs)

Epoch 1/5

938/938 ————— 2s 2ms/step - accuracy: 0.8590 - loss: 0.4948

Epoch 2/5

938/938 ————— 2s 2ms/step - accuracy: 0.9644 - loss: 0.1203

Epoch 3/5
938/938 ————— 2s 2ms/step - accuracy: 0.9762 - loss: 0.0805
Epoch 4/5
938/938 ————— 2s 2ms/step - accuracy: 0.9821 - loss: 0.0606
Epoch 5/5
938/938 ————— 2s 2ms/step - accuracy: 0.9855 - loss: 0.0477

Evaluating model performance on test data...
313/313 - 0s - 1ms/step - accuracy: 0.9708 - loss: 0.0973

Final Test Accuracy: 97.08%

In [7]:

```
!pip install --user tensorflow
```

```
Requirement already satisfied: tensorflow in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (2.20.0)
Requirement already satisfied: absl-py>=1.0.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (2.3.1)
Requirement already satisfied: wrapt>=1.11.0 in c:\programdata\anaconda3\lib\site-packages (from tensorflow) (1.12.1)
Requirement already satisfied: protobuf>=5.28.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (6.33.6)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (1.80.0)
Requirement already satisfied: libclang>=13.0.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (18.1.1)
Requirement already satisfied: requests<3,>=2.21.0 in c:\programdata\anaconda3\lib\site-packages (from tensorflow) (2.26.0)
Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (0.7.0)
Requirement already satisfied: google_pasta>=0.1.1 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (0.2.0)
Requirement already satisfied: opt_einsum>=2.3.2 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (3.4.0)
Requirement already satisfied: setuptools in c:\programdata\anaconda3\lib\site-packages (from tensorflow) (58.0.4)
Requirement already satisfied: tensorboard~2.20.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (2.20.0)
Requirement already satisfied: typing_extensions>=3.6.6 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (4.15.0)
Requirement already satisfied: h5py>=3.11.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (3.14.0)
Requirement already satisfied: numpy>=1.26.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (1.26.4)
Requirement already satisfied: termcolor>=1.1.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (3.1.0)
Requirement already satisfied: flatbuffers>=24.3.25 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (25.12.19)
Requirement already satisfied: six>=1.12.0 in c:\programdata\anaconda3\lib\site-packages (from tensorflow) (1.16.0)
Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (0.5.4)
Requirement already satisfied: keras>=3.10.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (3.10.0)
Requirement already satisfied: packaging in c:\programdata\anaconda3\lib\site-packages (from tensorflow) (21.0)
Requirement already satisfied: astunparse>=1.6.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorflow) (1.6.3)
Requirement already satisfied: wheel<1.0,>=0.23.0 in c:\programdata\anaconda3
```

\lib\site-packages (from astunparse>=1.6.0->tensorflow) (0.37.0)
 Requirement already satisfied: namex in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from keras>=3.10.0->tensorflow) (0.1.0)
 Requirement already satisfied: optree in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from keras>=3.10.0->tensorflow) (0.19.1)
 Requirement already satisfied: rich in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from keras>=3.10.0->tensorflow) (15.0.0)
 Requirement already satisfied: charset-normalizer~=2.0.0 in c:\programdata\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (2.0.4)
 Requirement already satisfied: urllib3<1.27,>=1.21.1 in c:\programdata\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (1.26.7)
 Requirement already satisfied: idna<4,>=2.5 in c:\programdata\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (3.2)
 Requirement already satisfied: certifi>=2017.4.17 in c:\programdata\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (2021.10.8)
 Requirement already satisfied: pillow in c:\programdata\anaconda3\lib\site-packages (from tensorboard~=2.20.0->tensorflow) (8.4.0)
 Requirement already satisfied: markdown>=2.6.8 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorboard~=2.20.0->tensorflow) (3.9)
 Requirement already satisfied: werkzeug>=1.0.1 in c:\programdata\anaconda3\lib\site-packages (from tensorboard~=2.20.0->tensorflow) (2.0.2)
 Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from tensorboard~=2.20.0->tensorflow) (0.7.2)
 Requirement already satisfied: importlib-metadata>=4.4 in c:\programdata\anaconda3\lib\site-packages (from markdown>=2.6.8->tensorboard~=2.20.0->tensorflow) (4.8.1)
 Requirement already satisfied: zipp>=0.5 in c:\programdata\anaconda3\lib\site-packages (from importlib-metadata>=4.4->markdown>=2.6.8->tensorboard~=2.20.0->tensorflow) (3.6.0)
 Requirement already satisfied: pyparsing>=2.0.2 in c:\programdata\anaconda3\lib\site-packages (from packaging->tensorflow) (3.0.4)
 Requirement already satisfied: pygments<3.0.0,>=2.13.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from rich->keras>=3.10.0->tensorflow) (2.20.0)
 Requirement already satisfied: markdown-it-py>=2.2.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from rich->keras>=3.10.0->tensorflow) (3.0.0)
 Requirement already satisfied: mdurl~=0.1 in c:\programdata\anaconda3\lib\site-packages (from markdown-it-py>=2.2.0->rich->keras>=3.10.0->tensorflow) (0.1.2)

In [8]:

```

import tensorflow as tf
from tensorflow.keras import layers, models

# 1. Load and preprocess the MNIST dataset
print("Loading MNIST dataset...")
mnist = tf.keras.datasets.mnist
(X_train, y_train), (X_test, y_test) = mnist.load_data()

# Normalize pixel values to be between 0 and 1
X_train, X_test = X_train / 255.0, X_test / 255.0

# 2. Build the Deep Neural Network (DNN) Architecture
print("Building the model architecture...")
model = models.Sequential([
    layers.Flatten(input_shape=(28, 28)),      # Flattens the 28x28 images into
    layers.Dense(128, activation='relu'),      # Hidden Layer 1 (128 neurons)
    layers.Dense(64, activation='relu'),       # Hidden Layer 2 (64 neurons)
    layers.Dense(10, activation='softmax')     # Output Layer (10 classes for
])
  
```

```

# 3. Compile the model
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

# 4. Train the model
print("Training the Deep Neural Network on MNIST...")
model.fit(X_train, y_train, epochs=5, batch_size=64)

# 5. Evaluate the model on test data
print("\nEvaluating model performance on test data...")
test_loss, test_acc = model.evaluate(X_test, y_test, verbose=2)
print(f"\nFinal Test Accuracy: {test_acc * 100:.2f}%")

```

Loading MNIST dataset...

Building the model architecture...

Training the Deep Neural Network on MNIST...

Epoch 1/5

938/938 ————— 2s 2ms/step - accuracy: 0.8550 - loss: 0.5040

Epoch 2/5

938/938 ————— 2s 2ms/step - accuracy: 0.9638 - loss: 0.1213

Epoch 3/5

938/938 ————— 2s 2ms/step - accuracy: 0.9758 - loss: 0.0783

Epoch 4/5

938/938 ————— 2s 2ms/step - accuracy: 0.9818 - loss: 0.0576

Epoch 5/5

938/938 ————— 1s 2ms/step - accuracy: 0.9866 - loss: 0.0415

Evaluating model performance on test data...

313/313 - 0s - 1ms/step - accuracy: 0.9719 - loss: 0.0911

Final Test Accuracy: 97.19%

In [9]:

```
!pip install --user "numpy<2.0.0" "scipy>=1.10.0" --upgrade
```

Requirement already satisfied: numpy<2.0.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (1.26.4)

Requirement already satisfied: scipy>=1.10.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (1.13.1)

In [10]:

```

import tensorflow as tf
from tensorflow.keras import layers, models

# 1. Load and preprocess the MNIST dataset
print("Loading MNIST dataset...")
mnist = tf.keras.datasets.mnist
(X_train, y_train), (X_test, y_test) = mnist.load_data()

X_train, X_test = X_train / 255.0, X_test / 255.0

# 2. Build the Deep Neural Network (DNN) Architecture
model = models.Sequential([
    layers.Flatten(input_shape=(28, 28)),
    layers.Dense(128, activation='relu'),
    layers.Dense(64, activation='relu'),
    layers.Dense(10, activation='softmax')
])

# 3. Compile the model
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

```

```

# 4. Train the model
print("Training the Deep Neural Network on MNIST...")
model.fit(X_train, y_train, epochs=5, batch_size=64)

# 5. Evaluate the model on test data
test_loss, test_acc = model.evaluate(X_test, y_test, verbose=2)
print(f"\nFinal Test Accuracy: {test_acc * 100:.2f}%")

```

Loading MNIST dataset...

Training the Deep Neural Network on MNIST...

Epoch 1/5

938/938 ————— 2s 2ms/step - accuracy: 0.8616 - loss: 0.4871

Epoch 2/5

938/938 ————— 2s 2ms/step - accuracy: 0.9620 - loss: 0.1267

Epoch 3/5

938/938 ————— 2s 2ms/step - accuracy: 0.9747 - loss: 0.0849

Epoch 4/5

938/938 ————— 2s 2ms/step - accuracy: 0.9811 - loss: 0.0627

Epoch 5/5

938/938 ————— 2s 2ms/step - accuracy: 0.9844 - loss: 0.0508

313/313 - 0s - 1ms/step - accuracy: 0.9733 - loss: 0.0908

Final Test Accuracy: 97.33%

In [11]: `!pip install "numpy<2" --upgrade`

Requirement already satisfied: numpy<2 in c:\users\ch.sc.u4aie24057\appdata\r
oaming\python\python39\site-packages (1.26.4)

In [12]:

```

import tensorflow as tf
from tensorflow.keras import layers, models

# 1. Load and preprocess the MNIST dataset
print("Loading MNIST dataset...")
mnist = tf.keras.datasets.mnist
(X_train, y_train), (X_test, y_test) = mnist.load_data()

X_train, X_test = X_train / 255.0, X_test / 255.0

# 2. Build the Deep Neural Network (DNN) Architecture
model = models.Sequential([
    layers.Flatten(input_shape=(28, 28)),
    layers.Dense(128, activation='relu'),
    layers.Dense(64, activation='relu'),
    layers.Dense(10, activation='softmax')
])

# 3. Compile the model
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

# 4. Train the model
print("Training the Deep Neural Network on MNIST...")
model.fit(X_train, y_train, epochs=5, batch_size=64)

# 5. Evaluate the model on test data
test_loss, test_acc = model.evaluate(X_test, y_test, verbose=2)
print(f"\nFinal Test Accuracy: {test_acc * 100:.2f}%")

```

Loading MNIST dataset...

Training the Deep Neural Network on MNIST...

Epoch 1/5

938/938 ————— 2s 2ms/step - accuracy: 0.8622 - loss: 0.4908
Epoch 2/5
938/938 ————— 2s 2ms/step - accuracy: 0.9630 - loss: 0.1240
Epoch 3/5
938/938 ————— 2s 2ms/step - accuracy: 0.9740 - loss: 0.0830
Epoch 4/5
938/938 ————— 1s 2ms/step - accuracy: 0.9801 - loss: 0.0640
Epoch 5/5
938/938 ————— 2s 2ms/step - accuracy: 0.9857 - loss: 0.0455
313/313 - 0s - 1ms/step - accuracy: 0.9707 - loss: 0.0879

Final Test Accuracy: 97.07%

In [14]: `!pip install --user torch torchvision`

```
Collecting torch
  Using cached torch-2.8.0-cp39-cp39-win_amd64.whl (241.2 MB)
Collecting torchvision
  Using cached torchvision-0.23.0-cp39-cp39-win_amd64.whl (1.6 MB)
Requirement already satisfied: jinja2 in c:\programdata\anaconda3\lib\site-packages (from torch) (2.11.3)
Collecting sympy<=1.13.3
  Using cached sympy-1.14.0-py3-none-any.whl (6.3 MB)
Requirement already satisfied: fsspec in c:\programdata\anaconda3\lib\site-packages (from torch) (2021.10.1)
Requirement already satisfied: filelock in c:\programdata\anaconda3\lib\site-packages (from torch) (3.3.1)
Requirement already satisfied: typing-extensions>=4.10.0 in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from torch) (4.15.0)
Requirement already satisfied: networkx in c:\programdata\anaconda3\lib\site-packages (from torch) (2.6.3)
Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in c:\programdata\anaconda3\lib\site-packages (from torchvision) (8.4.0)
Requirement already satisfied: numpy in c:\users\ch.sc.u4aie24057\appdata\roaming\python\python39\site-packages (from torchvision) (1.26.4)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in c:\programdata\anaconda3\lib\site-packages (from sympy<=1.13.3->torch) (1.2.1)
Requirement already satisfied: MarkupSafe>=0.23 in c:\programdata\anaconda3\lib\site-packages (from jinja2->torch) (1.1.1)
Installing collected packages: sympy, torch, torchvision
Successfully installed sympy-1.14.0 torch-2.8.0 torchvision-0.23.0

WARNING: The script isympy.exe is installed in 'C:\Users\ch.sc.u4aie24057\AppData\Roaming\Python\Python39\Scripts' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

WARNING: The scripts torchfrtrace.exe and torchrun.exe are installed in 'C:\Users\ch.sc.u4aie24057\AppData\Roaming\Python\Python39\Scripts' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
```

In [15]:

```
import torch
import torch.nn as nn
import torch.optim as optim
from torchvision import datasets, transforms
from torch.utils.data import DataLoader

# 1. Transform and Load the MNIST Dataset
transform = transforms.Compose([
    transforms.ToTensor(),
    transforms.Normalize((0.1307,), (0.3081,))
])
```



```

print("Loading MNIST dataset for PyTorch...")
train_dataset = datasets.MNIST(root='./data', train=True, download=True, tran
test_dataset = datasets.MNIST(root='./data', train=False, download=True, tran

train_loader = DataLoader(train_dataset, batch_size=64, shuffle=True)
test_loader = DataLoader(test_dataset, batch_size=1000, shuffle=False)

# 2. Define the Deep Neural Network (DNN) Architecture
class MNIST_DNN(nn.Module):
    def __init__(self):
        super(MNIST_DNN, self).__init__()
        self.flatten = nn.Flatten()
        self.fc1 = nn.Linear(28 * 28, 128) # Hidden Layer 1 (128 neurons)
        self.relu1 = nn.ReLU()
        self.fc2 = nn.Linear(128, 64)      # Hidden Layer 2 (64 neurons)
        self.relu2 = nn.ReLU()
        self.fc3 = nn.Linear(64, 10)      # Output Layer (10 digit classes)

    def forward(self, x):
        x = self.flatten(x)
        x = self.relu1(self.fc1(x))
        x = self.relu2(self.fc2(x))
        x = self.fc3(x)
        return x

# Instantiate the model
model = MNIST_DNN()

# 3. Define Loss Function and Optimizer
criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(model.parameters(), lr=0.001)

# 4. Train the Model
print("\nTraining the Deep Neural Network using PyTorch...")
model.train()
for epoch in range(1, 6): # 5 Epochs
    correct = 0
    total = 0
    running_loss = 0.0

    for batch_idx, (data, target) in enumerate(train_loader):
        optimizer.zero_grad()
        output = model(data)
        loss = criterion(output, target)
        loss.backward()
        optimizer.step()

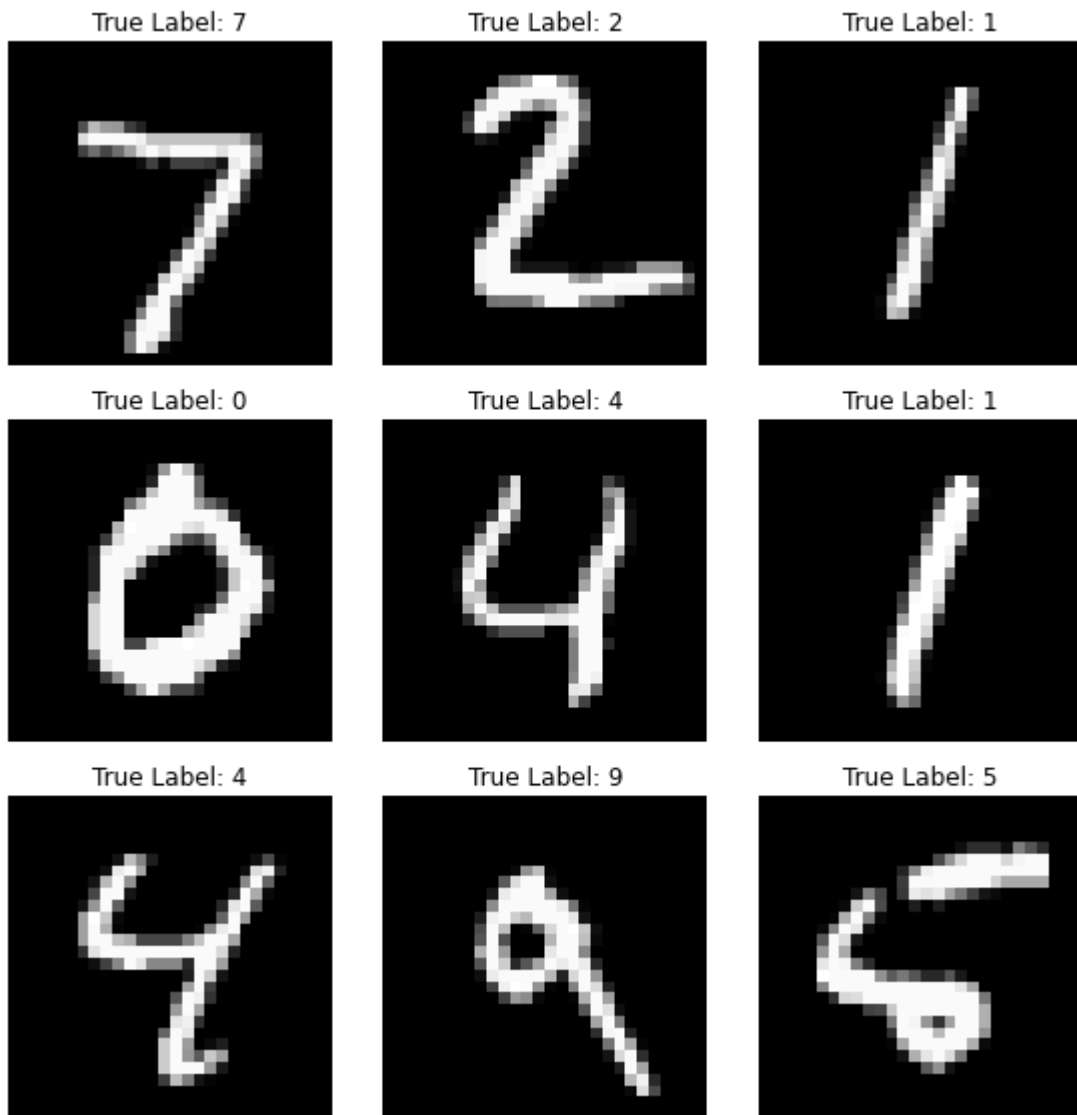
        running_loss += loss.item()
        _, predicted = torch.max(output.data, 1)
        total += target.size(0)
        correct += (predicted == target).sum().item()

    epoch_loss = running_loss / len(train_loader)
    epoch_acc = (correct / total) * 100
    print(f"Epoch {epoch}/5 -> Loss: {epoch_loss:.4f} | Accuracy: {epoch_acc:.4f}")

# 5. Evaluate the Model on Test Data
print("\nEvaluating PyTorch model performance on test data...")
model.eval()
correct = 0

with torch.no_grad():
    for data, target in test_loader:
        output = model(data)

```

In [18]:

```
import tensorflow as tf
from tensorflow.keras import layers, models
import matplotlib.pyplot as plt

# 1. Load data
mnist = tf.keras.datasets.mnist
(X_train, y_train), (X_test, y_test) = mnist.load_data()
X_train, X_test = X_train / 255.0, X_test / 255.0

# 2. Build model
model = models.Sequential([
    layers.Flatten(input_shape=(28, 28)),
    layers.Dense(128, activation='relu'),
    layers.Dense(64, activation='relu'),
    layers.Dense(10, activation='softmax')
])

model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

# 3. Train and SAVE the history of training
print("Training model and saving history...")
history = model.fit(X_train, y_train, epochs=5, batch_size=64, validation_dat

# 4. PLOT THE GRAPHS
plt.figure(figsize=(12, 4))
```

```

# Plot Accuracy Graph
plt.subplot(1, 2, 1)
plt.plot(history.history['accuracy'], label='Train Accuracy', color='blue', m
plt.plot(history.history['val_accuracy'], label='Test Accuracy', color='orang
plt.title('Model Accuracy Curve')
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.legend()
plt.grid(True)

# Plot Loss Graph
plt.subplot(1, 2, 2)
plt.plot(history.history['loss'], label='Train Loss', color='red', marker='o'
plt.plot(history.history['val_loss'], label='Test Loss', color='purple', mark
plt.title('Model Loss Curve')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.legend()
plt.grid(True)

plt.tight_layout()
plt.show()

```

C:\Users\ch.sc.u4aie24057\AppData\Roaming\Python\Python39\site-packages\keras\src\layers\reshaping\flatten.py:37: UserWarning: Do not pass an `input\_shape` or `input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

```
super().__init__(**kwargs)
```

Training model and saving history...

Epoch 1/5

938/938 ————— 2s 2ms/step - accuracy: 0.8606 - loss: 0.4896
- val_accuracy: 0.9564 - val_loss: 0.1395

Epoch 2/5

938/938 ————— 2s 2ms/step - accuracy: 0.9645 - loss: 0.1206
- val_accuracy: 0.9697 - val_loss: 0.0920

Epoch 3/5

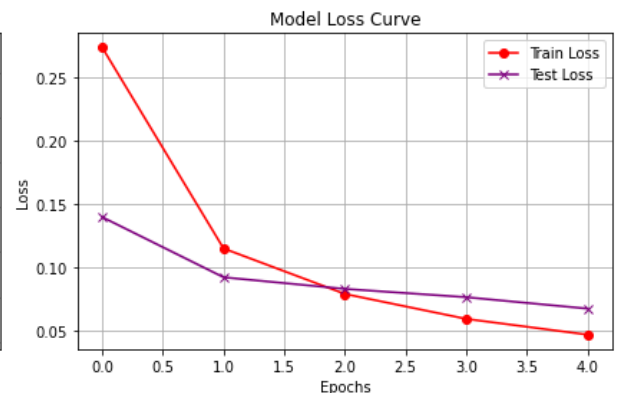
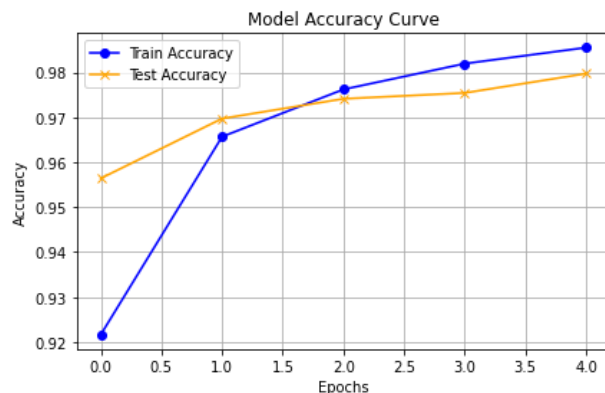
938/938 ————— 2s 2ms/step - accuracy: 0.9760 - loss: 0.0812
- val_accuracy: 0.9741 - val_loss: 0.0829

Epoch 4/5

938/938 ————— 2s 2ms/step - accuracy: 0.9823 - loss: 0.0584
- val_accuracy: 0.9754 - val_loss: 0.0764

Epoch 5/5

938/938 ————— 2s 2ms/step - accuracy: 0.9867 - loss: 0.0448
- val_accuracy: 0.9797 - val_loss: 0.0674



In []: